Ex: $x[n] = \alpha^n u[n]$, $|\alpha| < 1$, $y[n] = ?$

$h[n] = \beta^n u[n]$, $|\beta| < 1$

$$y[n] = x[n] * h[n] = \mathcal{F}^{-1}\{Y(e^{j\omega}) = X(e^{j\omega}) H(e^{j\omega})\}$$

Reminder: $\alpha^n u[n] \xleftrightarrow{\mathcal{F}} \frac{1}{1-\alpha e^{-j\omega}}$

$$Y(e^{j\omega}) = \frac{1}{(1-\alpha e^{-j\omega})(1-\beta e^{-j\omega})} = \frac{A}{1-\alpha e^{-j\omega}} + \frac{B}{1-\beta e^{-j\omega}}$$

$\updownarrow \mathcal{F}^{-1}$

$$y[n] = A\alpha^n u[n] - B\beta^n u[n]$$

$$A = \frac{\alpha}{\alpha-\beta} \qquad B = -\frac{\beta}{\alpha-\beta}$$

$$y[n] = \frac{1}{\alpha-\beta}\left[\alpha\,\alpha^n u[n] - \beta\,\beta^n u[n]\right] = \frac{1}{\alpha-\beta}\left(\alpha^{n+1} - \beta^{n+1}\right)u[n]$$

Ex: $\alpha = \beta$ ise $y[n] = ?$

$$Y(e^{j\omega}) = \frac{1}{(1-\alpha e^{-j\omega})^2}$$

Reminder: $x[n] = \alpha^n u[n] \xleftrightarrow{\mathcal{F}} X(e^{j\omega}) = \frac{1}{1-\alpha e^{-j\omega}}$

$$n\,x[n] \xleftrightarrow{\mathcal{F}} j\frac{dX(e^{j\omega})}{d\omega}$$

$$x[n-n_0] \xleftrightarrow{\mathcal{F}} e^{-j\omega n_0} X(e^{j\omega})$$

$$\frac{dX(e^{j\omega})}{d\omega} = -j\alpha e^{-j\omega}\frac{1}{(1-\alpha e^{-j\omega})^2} = -j\alpha e^{-j\omega} Y(e^{j\omega})$$

$$Y(e^{j\omega}) = \frac{1}{\alpha} e^{j\omega}\, j\frac{dX(e^{j\omega})}{d\omega} \xleftrightarrow{\mathcal{F}^{-1}} y[n] = \frac{1}{\alpha}(n+1)\alpha^{n+1}u[n+1] = (n+1)\alpha^n u[n]$$

$\frac{dX(e^{j\omega})}{d\omega}$: Frekansta türev

$e^{j\omega}$: Zamanda öteleme

$n\alpha^n u[n]$

$y[-1] = 0$

$$(n+1)\alpha^n u[n] \xleftrightarrow{\mathcal{F}} \frac{1}{(1-\alpha e^{-j\omega})^2}$$

Ex: $y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = 2x[n]$ ise $h[n] = ?$

$$Y(e^{j\omega}) - \frac{3}{4}e^{-j\omega}Y(e^{j\omega}) + \frac{1}{8}e^{-j2\omega}Y(e^{j\omega}) = 2X(e^{j\omega})$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{2}{1 - \frac{3}{4}e^{-j\omega} + \frac{1}{8}e^{-j2\omega}} = \frac{A}{1 - \frac{1}{2}e^{-j\omega}} + \frac{B}{1 - \frac{1}{4}e^{-j\omega}}$$

$A = 4$, $B = -2$

$\updownarrow \mathcal{F}^{-1}$

$$h[n] = 4\left(\frac{1}{2}\right)^n u[n] - 2\left(\frac{1}{4}\right)^n u[n]$$